EXPERIMENT REPORT

1. Executive Summary

This report presents the experimental results for the warehouse-demand forecasting system built around a first-principles single-head self-attention mechanism. The experiment proceeds from a controlled toy learning task to the full synthetic warehouse forecasting problem, then tests the fixed model under distribution shift and investigates the largest individual forecast errors.

The base warehouse test contains 1,311 forecast targets. The attention model produced MAE 18.5462, RMSE 26.5753, and MAPE 5.30%. Persistence produced MAE 18.5331, RMSE 30.0543, and MAPE 5.34%. The results therefore show very similar average absolute error on this synthetic environment, while the attention model has a lower RMSE, indicating a different distribution of errors with fewer or smaller very-large misses in aggregate.

The distribution-shift experiments were performed without retraining. A +40 level shift changed attention MAE to 25.3494, while doubling the noise standard deviation from 12 to 24 changed attention MAE to 30.1605. The fixed model therefore changes materially when the data-generating conditions change.

The failure investigation found 1,311 test forecasts, a largest absolute error of 211.5344, and a mean target jump of 49.5123 among the top 20 errors. The largest misses were associated with large one-hour demand changes, consistent with the designed rare-event component of the synthetic generator.

2. Experimental Design

The synthetic warehouse environment contains 365 days of hourly demand, giving 8,760 observations. The forecasting problem uses the previous 24 hourly demand values to predict the next hour.

The supervised dataset contains 8,736 valid 24-hour input/next-hour target examples. The chronological target split is 6,115 training examples, 1,310 validation examples, and 1,311 test examples.

The model is a small, single-head architecture: scalar demand is projected to d_model = 4, sinusoidal positional encoding is added, first-principles attention is applied, and a linear head maps the final-position representation to the predicted demand.

Training uses Adam with learning rate 0.01, mean squared error loss, 200 epochs, and seed 42. Standardization parameters are computed from training data only. The best validation checkpoint is restored for final test evaluation.

Three simple baselines are evaluated using exactly the same test targets: persistence, previous-day same-hour, and 24-hour mean.

3. Toy Learning Dynamics

Before the warehouse task, the attention implementation was tested on a deterministic next-value problem with 512 samples and sequence length 8. The target is y = 0.60 x[-1] + 0.25 x[-2] + 0.10 x[-3] + 0.05 x[-4]. This makes the relevant temporal structure known in advance.

Two otherwise identical models were trained: scaled attention using QK^T / sqrt(d_k), and unscaled attention using QK^T. Only the score-scaling choice was changed.

In the recorded run, scaled attention reduced MSE from 1.977446 at epoch 1 to 0.088134 at epoch 100. Unscaled attention reduced MSE from 2.267351 to 0.087712. Both variants learned substantially.

The final losses are close, so this run does not establish that scaling always produces a lower final loss. Its value is diagnostic: the controlled experiment shows that the implementation can learn the known temporal rule and that scaling changes the optimization trajectory.

4. Warehouse Forecasting Results

The attention model was evaluated against three interpretable forecasting rules on the same chronological test set. Results are shown below.

Model	MAE	RMSE	MAPE
Persistence	18.5331	30.0543	5.34%
Previous-day same-hour	24.7375	36.0324	7.25%
24-hour mean	33.8771	41.5632	9.94%
Attention	18.5462	26.5753	5.30%

5. Interpretation of the Base Test

The persistence baseline has the smallest MAE by a very small margin on this particular synthetic test set. This is consistent with the strong hour-to-hour temporal correlation created by the data-generating process.

The attention model has a slightly larger MAE than persistence but a smaller RMSE. Because RMSE weights large errors more strongly, the difference suggests that the two methods distribute their errors differently, with the attention model producing a lower aggregate contribution from large misses.

The previous-day same-hour baseline uses the explicit 24-hour seasonal structure but does not match the recent level as directly as persistence. The 24-hour mean smooths the sequence but removes much of the local state information.

These are observations for this synthetic environment and this fixed experimental configuration. They are not claims about forecasting performance on real warehouse data.

6. Distribution Shift

Two changes were then applied to the test environment, without retraining: a +40 demand-level shift and an increase of the ordinary noise standard deviation from 12 to 24.

The level-shift experiment tests sensitivity to a changed demand level while preserving the basic temporal structure. The noise experiment tests sensitivity when the target becomes less predictable from the historical signal.

Scenario	Model	MAE	RMSE	MAPE
Original test	Persistence	18.5331	30.0543	5.34%
Original test	Previous-day same-hour	24.7375	36.0324	7.25%

Original test	24-hour mean	33.8771	41.5632	9.94%
Original test	Attention	18.5462	26.5753	5.30%
Level shift +40	Persistence	18.5636	30.0817	4.78%
Level shift +40	Previous-day same-hour	25.0080	36.4000	6.54%
Level shift +40	24-hour mean	33.8002	41.5871	8.84%
Level shift +40	Attention	25.3494	34.4030	6.29%
Noise std 24	Persistence	33.5483	44.9858	9.84%
Noise std 24	Previous-day same-hour	37.6624	49.2329	11.08%
Noise std 24	24-hour mean	37.9344	47.3996	11.26%
Noise std 24	Attention	30.1605	39.1146	8.85%

8. Interpretation of the Generalization Experiment

Under the +40 level shift, the attention model's MAE increases from 18.5462 to 25.3494. Its RMSE increases from 26.5753 to 34.4030. This shows that the fixed model is sensitive to a shift in demand level.

Under the noise-standard-deviation shift, attention MAE increases from 18.5462 to 30.1605 and RMSE increases from 26.5753 to 39.1146. The larger degradation is consistent with the intended mechanism of the experiment: more observation noise reduces the information available in the past 24 hours.

The baselines also change under these shifts. This is important because the generalization test is not interpreted only as an attention-model score; the same changed environments are applied to all reference methods so that the response of the complete forecasting system can be compared.

9. Failure Investigation

The failure investigation keeps the model fixed and examines individual test cases rather than training another model. For each forecast, absolute error $e_t = |y_t - \hat{y}_t|$ is calculated and the 20 largest errors are retained.

For every high-error case, the previous observed demand D_t and target jump $y_{(t+1)} - D_t$ are recorded. A large target jump is a useful diagnostic because the synthetic generator deliberately contains rare spikes and drops. It is not treated as proof that every error came from a specific injected event.

The largest five errors are shown below. The largest absolute error is 211.53 demand units, and all five largest misses have target jumps close to 189–199 units. This provides concrete evidence that abrupt one-hour changes are a major source of difficult forecasts in the generated environment.

The script also averages the final-query attention weights across all test examples. The final query corresponds to the last input position because the forecast is produced from output[:, -1, :]. The averaged weights describe where the model places attention on average; they are not interpreted as causal importance.

Target timestamp	Actual	Prediction	Absolute error	Target jump
2025-11-27 08:00	587.19	375.66	211.53	189.96
2025-12-30 02:00	586.98	376.24	210.74	199.09
2025-12-12 18:00	523.57	333.75	189.82	198.95
2025-12-01 18:00	495.67	310.85	184.82	188.77
2025-12-24 15:00	505.86	323.40	182.47	189.10

10. Reproducibility and Final QA

Reproducibility was checked at two levels: synthetic data generation and neural-network training. The same 365-day generator with seed 42 was run twice and the resulting tables and demand-vector SHA-256 hashes were compared. The attention model was also trained twice with the same seed and configuration, and the loss history, parameter tensors, training mean, and training standard deviation were compared exactly.

The final project QA separately checked the expected project structure, generated outputs, result files, and the complete test suite. The recorded final QA status is PASS.

11. Overall Experimental Findings

The experiments demonstrate a complete chain from controlled mathematical verification to a forecasting task with explicit baselines, robustness testing, and failure analysis.

The base test shows that simple persistence is already highly competitive because the synthetic process is strongly time-correlated. The attention model provides a similar MAE while lowering RMSE relative to persistence on this test set.

The generalization experiments show that the fixed attention model is sensitive to changes in the data-generating process, especially increased noise. The failure analysis shows that the largest individual errors occur when demand changes sharply from one hour to the next.

Taken together, the results support a more specific conclusion than simply saying that attention 'works': the implemented attention mechanism learns the controlled temporal task, gives competitive forecasts on the designed warehouse environment, changes its behavior under controlled shifts, and has identifiable failure cases associated with abrupt demand changes.

12. Limitations of the Experiment

The warehouse environment is synthetic. The results therefore measure behavior under the chosen data-generating process rather than real operational demand.

The attention architecture is intentionally small (d_model = 4, one head) and is designed to expose the core mechanism rather than maximize model capacity.

The distribution-shift experiments cover two controlled shifts only: a level shift and increased ordinary noise. They do not cover every possible real-world shift.